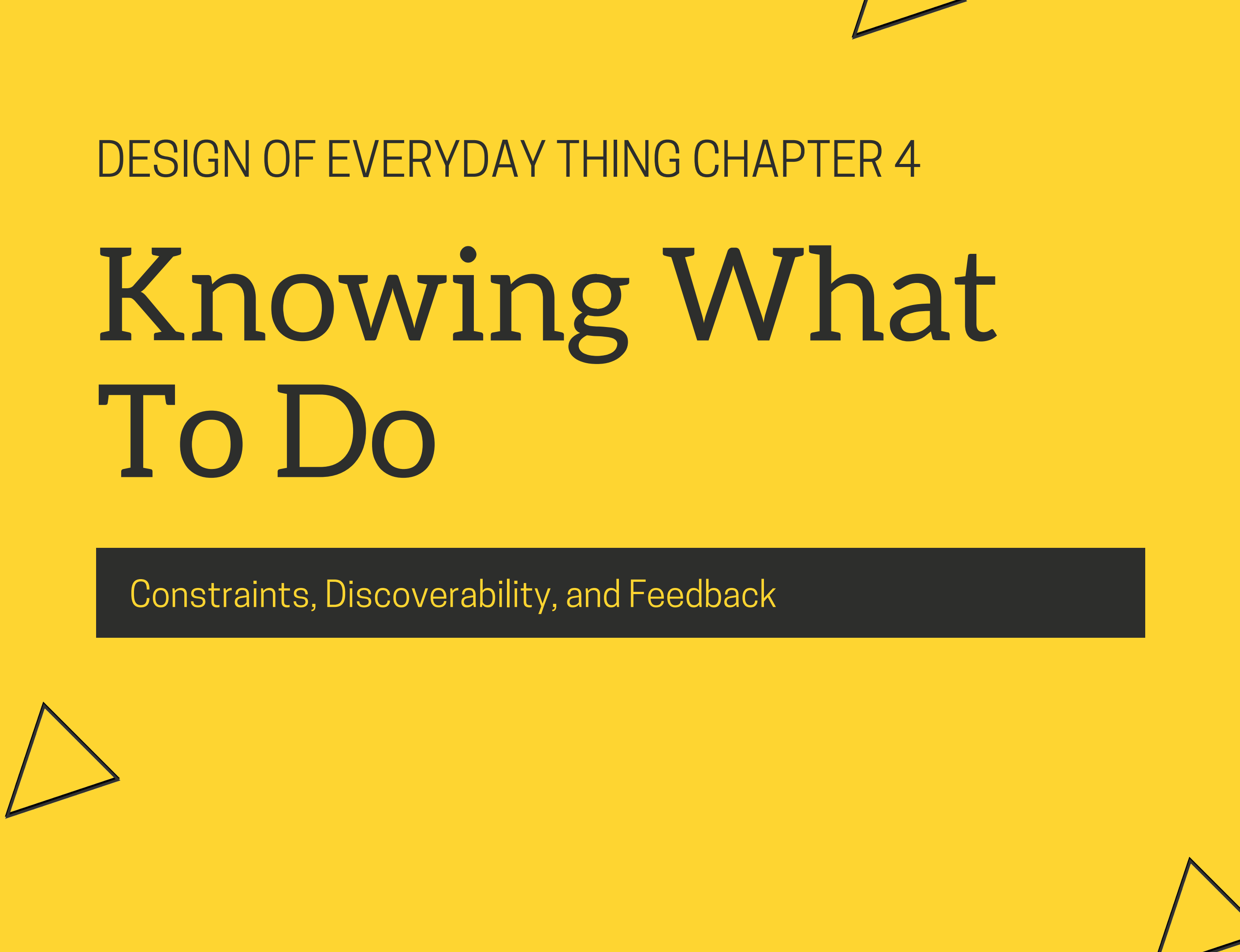




DESIGN OF EVERYDAY THING CHAPTER 4

Knowing What To Do

Constraints, Discoverability, and Feedback





Introduction

How do we interact with unfamiliar objects?

- We do this by combining two types of knowledge: Knowledge in the world and knowledge in our head.

Knowledge in the world:

- Refers to the information we can perceive directly from our environment, such as signifiers, affordance, and constraints. This knowledge is readily available and doesn't require learning.

Knowledge in the head:

- The knowledge we have stored in our memory, including our understanding of cultural norms, language, and past experiences. This knowledge requires learning and can be more efficient to use, but it's not always readily available.

Chapter's main focus on "knowledge in the world":

- It explores how designers can embed clues and constraints into the design of objects to guide users and help them understand how to use them, even if they have never encountered them before.

THE POWER OF CONSTRAINTS

- Constraints are powerful design elements that limit the possible actions a user can take. By reducing the number of choices, constraints guide the user toward the correct action and make it more difficult to make mistakes.
- The Lego Motorcycle example: Even without instructions, a child can assemble a Lego motorcycle because of the constraints of the pieces. The shapes of the blocks only allow them to be connected in specific ways, making the assembly process a puzzle with clear solution.
- Constraints simplify decision making: They reduce the need for conscious thought and allow us to act quickly and effectively.



FOUR KINDS OF CONSTRAINTS

- **Physical :** These constraints are related to the physical properties of objects, such as their size, shape, and movement. **Example** USB Plug fits only one way.
- **Cultural :** Social norms and learned rules. **Example:** Red= Stop Green= GO
- **Semantic :** Meaning and Context suggest correct use **Example:** Bike seat faces forward
- **Logical :** Based on ability to use reason and logic to understand how system works. Process of elimination. **Example:** Only one puzzle piece left → only one spot it fits.

LEGACY PROBLEM

- **Legacy problem** is the difficulty of changing a widely adopted standard, even if a better design exists. This is because the cost of changing can be very high, and it can be disruptive to users who are already familiar with the old standard.

Examples:

- **QWERTY keyboard:** This layout was designed to slow down typists to prevent the keys on mechanical typewriters from jamming. While more efficient layouts exist, QWERTY remains the standard because so many people have learned to use it.
- **Cylindrical batteries:** The symmetrical design of cylindrical batteries makes it easy to insert them incorrectly. A better design would use physical constraints to prevent this, but the cylindrical battery is so widely used that it is difficult to change.



DISCOVERABILITY

A good design tells you what actions are possible

- **Signifiers (labels, shapes, sounds) help reveal functions.**
- **Poor discoverability = users guess or make mistakes.**
- **💡 Example: A faucet with identical knobs but unclear hot/cold mapping → trial and error.**

CONVENTIONS AND CULTURE

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- **Conventions** are a type of cultural constraint that helps us navigate new situations by providing a set of shared expectations for behavior. They are essential for a functioning society.
 - **But conventions can vary across cultures:**
 - **UK: flashing headlights → “You go first.”**
 - **Mexico: flashing headlights → “I am going first.”**
 - **Designers must consider global context or stick to universally accepted norms.**



FORCING FUNCTIONS

Forcing functions are a powerful type of physical constraint that makes it impossible to perform an action incorrectly. They are used in situations where an error could have serious consequences.

- The purpose of forcing functions is to prevent errors before they happen. They are a proactive approach to safety and usability.
- There are three main types of forcing functions:
- **Interlocks:** Prevent an action from happening in the wrong sequence.
- **Lock-ins:** Keep an operation active, preventing it from being stopped prematurely.
- **Lockouts:** Prevent a user from entering a dangerous area or performing a dangerous action.



INTERLOCKS, LOCK-INS, AND LOCKOUTS

- **Interlocks:** ▪ **Microwave ovens:** A microwave will not start if the door is open. This is an interlock that prevents the user from being exposed to microwave radiation.
- **"Dead man's switch":** A switch that must be held down for a machine to operate. If the operator lets go of the switch, the machine stops.
- **Lock-ins:** ▪ **"Do you want to save your changes?"** dialog box: This is a lock-in that prevents you from closing a document without saving your work.
- **Confirming a purchase:** When you are about to make an online purchase, you are often asked to confirm your order. This is a lock-in that prevents you from making an accidental purchase.
- **Lockouts:** ▪ **Childproof caps:** These are a type of lockout that prevents children from opening bottles of medicine or dangerous chemicals.
- **Safety guards on machinery:** These are physical barriers that prevent workers from coming into contact with dangerous moving parts.



SOUND AS A SIGNIFIER

Sound can be a powerful signifier, providing information and feedback to the user.

- **The click of a key:** The sound of a key turning in a lock provides feedback that the lock has been successfully operated.
- **The sound of a car's engine:** The sound of a car's engine provides information to pedestrians about the car's presence and speed.
- **The challenge of sound design:** The challenge for designers is to create sounds that are informative and helpful without being annoying or distracting.
- **The sound of electric cars:** Because electric cars are so quiet, they can be a danger to pedestrians and cyclists who may not hear them approaching. As a result, many electric cars are now designed to emit an artificial sound to alert people to their presence.

SKEUOMORPHISM

Skeuomorphism is the design practice of making new technologies look and feel like their older, more familiar counterparts. This can help to make new technologies feel less intimidating and easier to learn.

Examples of skeuomorphism:

- **Early automobiles:** Many early automobiles were designed to look like "horseless carriages," with features like a whip holder, even though there was no horse to whip.
- **Computer desktops:** The use of "folders" and "files" on a computer desktop is a skeuomorphic design that mimics the way we organize paper documents in a physical office.
- **Digital calendars:** Many digital calendars are designed to look like a physical desk calendar, with a leather-like border and the sound of a page turning when you switch to the next month.

THE PROBLEM WITH DOORS

- Doors are good example of a simple object that can be made confusing by poor design. When a door handle that can be both pushed and pulled, but it only opens in one direction, this can lead to frustration and errors. It is called “Norman Door” name after author.
- **The importance of signifiers:** A well-designed door uses clear “signifiers” to indicate how it should be used.
 - Example: A sign labeling that it should be pushed or pulled.
- **Good design is invisible:** When a door is well-designed, we don’t even have to think about how to use it. We just see the signifiers and act.



THE PROBLEM WITH SWITCHES

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- Rows of identical switches can be source of confusion and errors, especially when the relationship between the switches and the services they control is not clear. This is problem of “mapping.”
 - Mapping reffers to the relationship between a control, its movement, and the result of the movement.
 - Good mapping is natural and intuitive:
- Stove controls: A well-designed stove will have the controls

CONCLUSION

- Good design is about more than just aesthetics; it's about making products and systems that are easy to understand and use.
- By understanding and applying the principles of constraints, signifiers, and feedback, designers can create products that are both functional and enjoyable.
- The goal of good design is to make the user's experience as seamless and intuitive as possible. When a product is well-designed, we don't even have to think about how to use it; it just feels natural.

